IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 23 July 2024, accepted 12 August 2024, date of publication 27 August 2024, date of current version 18 September 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3450444

RESEARCH ARTICLE

An Ensemble Deep Learning Model for Oral Squamous Cell Carcinoma Detection Using Histopathological Image Analysis

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ABSTRACT Deep learning approaches for medical image analysis are widely applied for the recognition and classification of different kinds of cancer. In this study, histopathological images of oral cells are analyzed for the programmed recognition of Oral squamous cell carcinoma (OSCC) using the proposed framework. The suggested model applies transfer learning and ensemble learning in two phases. In the 1st phase, a few Convolutional neural network (CNN) models are considered through transfer learning applications for OSCC detection. In the 2nd phase, the ensemble model is constructed considering the best two pre-trained CNN from the 1st phase. The proposed classifier is compared with leading-edge models like Alexnet, Resnet50, Resnet101, Inception net, Xception net, and InceptionresnetV2. Results are analyzed to demonstrate the effectiveness of the suggested framework. A three-phase comparative analysis is considered. Firstly, various metrics including accuracy, recall, F-score, and precision are evaluated. Secondly, a graphical analysis using a loss and accuracy graph is performed. Lastly, the accuracy of the proposed classifier is compared with that of other models from existing literature. Following the three-stage performance evaluation, the proposed ensemble classifier exhibits enhanced performance with an accuracy of 97.88%.

INDEX TERMS Ensemble learning, histopathological oral cell images, InceptionresnetV2, oral squamous cell carcinoma, xception net.

I. INTRODUCTION

Oral cancer is a widespread and deadly disease globally. Within this group, oral squamous cell carcinoma (OSCC) signifies a broad range of malignancies originating in the mucous membranes of the mouth, encompassing over 90% of all oral malignancy occurrences [1], [2]. The World Health Organization reports an annual registration of 657,000 new cases and 330,000 deaths worldwide. Most cases of OSCC are recorded in South Asian countries; however, in India, one-third of cases are reported, while in Pakistan, OSCC is the first widespread cancer in males and the second-highest predominantly prevalent malignancy within the female population [3]. The main sign of oral malignancy is a sore in the mouth that doesn't go away and hurts when touched or bleeds. The sore or ulcer can appear in the inner part of lips, tongue, gums, or in the inner area of cheeks. Difficulty swallowing and chewing, swelling of the jaw, and persistent throat problems with difficulty in speaking are the common symptoms of oral cancer. When ulcers aren't treated, there's a bigger chance of getting oral cancer. The widespread menace factors of oral cancer are the unhygienic habit of tobacco chewing, alcohol consumption, exposure to human papilloma virus (HPV), genetic disorders, and geographical aspects. The OSCC has a poor survival rate of 50 percent. To address this challenge, timely identification of oral cancer is crucial for proper treatment and for improving survival rates by decreasing mortality cases [4], [5].

The associate editor coordinating the review of this manuscript and approving it for publication was Sandra Costanzo.

Currently, histopathological analysis under microscopic examination of biopsy or tissue of the affected oral cell by the histopathologist is the golden standard for oral cancer detection [6]. The primary standards utilized by histopathologists to detect cancer rely on parameters such as nuclear morphology, cellular distribution, size and shape of cells, and cellular growth patterns. Histopathologists need to analyze cells with necked eyes and interpret the visual information based on previous understanding of medical science. This approach to diagnosing cancer is time-intensive and vulnerable to inaccuracies, as it heavily hinges on the histopathologist's experience, workload, and judgment [7]. As a result, there is a great need for an efficient diagnostic tool that can assist histopathologists and doctors in identifying oral malignancies more efficiently.

In the modern era, due to the progress of computing technology and storage capacity, computer assistant diagnostic (CAD) technology become popular in medical fields. Moreover, in the area of histopathological image analysis, CAD systems are widely used. The histopathological images are generated using a digital scanner. The histopathology slides containing cells and tissue can be digitized by a whole slide digital scanner and stored as histopathological images. Histopathological images can be filtered rapidly and utilized by CAD technique for classification and detection of cancer. This results in reducing the burden of histopathologists.

Artificial intelligence (AI) has emerged as a pivotal player in enhancing medical diagnosis in the past few years. AI is a technique that mimics human behaviour for decision-making. One of the AI techniques, deep learning (DL), has exhibited noteworthy success, especially in the context of diagnosing cancer within pathological images. In recent years, based on DL, particularly using convolutional neural network (CNN), various CAD applications have been designed and constructed to tackle a range of cancer detection and classification tasks, spanning skin, breast, brain, liver, lung, prostate, and other type of cancers [8], [9], [10], [11]. However, from the literature, it is revealed that DL has been scarcely used for the detection of OSCC using histopathological image analysis.

In this work, a classification model based on transfer learning and deep ensemble learning has been proposed for oral cancer classification using histopathological images. Ensemble learning can augment the benefits of the DL approach by enhancing accuracy and generalization capabilities.

The subsequent sections of the document are arranged as outlined. The II part delves into related works. The research gap and key contributions of this research are discussed in the III part. In part IV, the methods used, namely transfer learning and ensemble learning concepts, are discussed, and the steps of the suggested ensemble approach are given. The methodology behind each step of the suggested model, such as the dataset used, preprocessing and augmentation techniques used, etc, are given in part V. In part VI, an experimental study is carried out; in part VII, results and discussion are mentioned; and finally, in part VIII, the conclusion and future scope are given.

II. RELATED WORK

In this segment, numerous related works for oral malignancy detection and classification are discussed and categorized into two distinct groups. The classification of oral malignancy utilizing machine learning (ML) methodologies is discussed in the first category, and in the second category, DL-dependent models are discussed.

A. CLASSIFICATION OF ORAL MALIGNANCY USING ML: A REVIEW

The predominant approach in research for the identification of oral malignancy involves methodology rooted in ML. Krishnan et al. [12] employed advanced signal processing techniques, including higher-order spectra and local binary pattern (LBP) features, extracted from histopathological oral images. These features were then provided into a support vector machine (SVM) classifier for the automated diagnosis of oral cancer. Krishnan et al. used 146 histopathological sample images with 69 textural features and investigated, that, the biorthogonal3.1 wavelet coefficient with local binary pattern and Gabor wavelet feature led to higher accuracy of 88.38% for classifying oral images using SVM classifier [13]. For the investigation of oral submucous fibrosis (OSF), Patra et al. [14] used wavelet-based multi-resolution technique and extracted 12 textural information of the spinous layer and used 6 significant features to train SVM and Bayesian classifier for discrimination of normal, OSF with abnormal cell growth and OSF without abnormal cell growth, also achieve an accuracy of 93.3% for Bayesian and 96.6% for SVM classifier. Thomas et al. [15] analyzed digital images of oral cancer. They extracted features using advanced methods like gray level co-occurrence matrix and gray level run length. Additionally, they considered intensity-based first-order features. The validation of the feature set is carried out with a back propagation-based artificial neural network. These features were then utilized to classify different types of oral malignancy. feature set. To increase the accuracy of classification they have used feature selection using box plot analysis. Prabhakar and Rajaguru [16] analyzed 75 oral cancer patients considering Tumour Node Metastasis variables and found out 100% accuracy in 1st stage, similarly 85.19%, 84.21%, and 94.1% in 2nd, 3rd and 4th stages respectively. Rahaman et al. [17] used a histogram and grey-level co-occurrence matrix for textural feature extraction from the histopathological image of oral cells using a linear support vector machine. Rahaman et al. [18] in another study used features such as shape, texture and color from the histopathological oral cell images and different classifiers are applied for cancer cell detection. The analysis concluded that the SVM and Linear Discriminant classifier produced optimal outcomes. In [19], [20], and [21] systematic reviews were carried out on several ML approaches used for the identification of oral malignancy. Though ML based approaches are widely used by researchers, but in the current generation, DL based approaches have earned massive popularity in the medical research field, specifically in cancer detection. Concerning this, in the next subsection, DL based models are addressed for oral cancer detection.

B. CLASSIFICATION OF ORAL MALIGNANCY USING DL: A REVIEW

DL techniques mainly CNN become revolutionary approaches in image analysis assignments. Various categories of malignant identification from images using DL based approach is the current research trend. In the DL approach, feature extraction and classification are carried out simultaneously and automatically, which makes this approach more popular.

Aubreville et al. [22] employed the DL method for the identification of oral malignancy using laser endomicroscopy images of the oral cell. Folmsbee et al. [23] employed a data-driven approach to classify digitally captured images of hematoxylin and eosin-stained oral specimens into seven discrete classes: adipose, mucosa, lymphocytes, blood, tumor, stroma, and keratin pearls. They utilized a hybrid methodology integrating Random Learning and Active Learning strategies via CNNs for the classification task and concluded that Active Learning's accuracy is better than Random Learning's accuracy. Ariji et al. [24] utilized CT scan images of the oral specimens and proposed five different learning models based on Alex Net. Finally, the authors have concluded that the results of DL models are better than the radiologist's performance. Similarly, Ariji et al. [25] also used CT scan images for the experiment and found, that the output of the DL model is comparable with the radiologist‘s output. Hyperspectral images are used by Jeyaraj and Nadar [26] for oral cancer detection using a regression-based partitioned CNN model. Similarly, Halicek et al. [27] also used hyperspectral images for oral malignancy identification using DL models and concluded that, hyperspectral images gave better performance as compared to proflavine dye and autofluorescence. Magnetic resonance imaging (MRI) images are used by Bhandary et al. [28]. They suggested a CNN model for the oral malignancy classification task and got an accuracy of 96.5%.

In the mentioned literature [22], [23], [24], [25], [26], [27], [28], oral cancer classification using varieties of DL models on various forms of images, such as endomicroscopy, digitized, CT scan, hyperspectral and MRI images of oral cavity are discussed. However, the next section of the literature study is focused on oral cancer detection using histopathological images.

Das et al. [29] suggested a CNN approach along with gober filter-trained random forest for the identification of oral malignant using the histopathological image also got an accuracy of 96.88%. Amin et al. [30] recommended a concatenated DL framework for histopathological oral image identification and recorded an accuracy of 96.66%. Das et al. [31] utilized four types of pre-initialized transfer learning frameworks, and also suggested a CNN approach, for the identification of four types of OSCC and got the best accuracy of 97.5% with the CNN model as compared to other transfer learning frameworks. Panigrahi and Swarnkar [32] proposed a modified Resnet model and achieved an accuracy of 97.59% and in [33] proposed a modified capsule network for oral malignancy detection using histopathological images and obtained an accuracy of 97.35%. Das et al. [34] suggested a CNN approach using histopathological oral cavity images and got an accuracy of 97.82%. Sukegawa et al. [35] uses VGG16, with a learning rate scheduler and SAM optimizer, for significantly improved pathologists' diagnostic accuracy.

Based on the literature reviewed, there is substantial scope for advancing oral cancer detection through the exploration of traditional ML, DL and ensemble modeling approaches. This literature study shows extensive use of techniques such as SVMs, Bayesian classifiers, and advanced feature extraction methods from various imaging modalities including histopathological images, CT scans, and MRI. DL methods, particularly CNNs, have shown promise in automatically extracting features and achieving high accuracy in classifying oral malignancies across different types of image data.

Even lots of research has been done using ML and DL approaches for oral malignant identification, however, relatively limited research carried out utilizing histopathological images to explore DL approaches for the automatic identification of oral malignancy.

This research suggested a competent ensemble model, that combines two pre-trained networks, namely InceptionresnetV2 and Xception net for the binary classification of oral cell histopathological images.

III. RESEARCH GAP AND CONTRIBUTION

In this section, the research gap and the key contribution of this study are discussed in two separate sections.

A. RESEARCH GAP AND MOTIVATION

Drawing from the scholarly work discussed in the previous section, it is noted that, though much research has been conducted for the automatic detection of oral cancer using ML and DL techniques, however, there are certain limitations in these approaches. Below the limitations of ML technique and DL techniques are discussed.

Traditional ML methods have been commonly applied in medical image analysis, including the classification of oral cancer. However, these approaches have drawbacks such as needing manual feature engineering, limitations in learning complex patterns, difficulties with large datasets, and struggles with handling complex relationships and generalizing to new data. These challenges encourage researchers to explore deep learning models.

Deep learning offers benefits like automatic extraction of features from data, scalability to large datasets, support transfer learning and the capability to learn intricate patterns directly from raw data. These advantages have the potential to improve the accuracy of diagnosing oral cancer.

It is understood from the literature that the dataset utilized for training the DL framework exhibits class imbalance issues. Consequently, while training DL models, proper data augmentation processes are necessary to overcome from the class imbalance issue while training the DL models.

Though CNN has generated abundant advancements in the area of image classification, it has certain challenges, such as training the preliminary layers, which is comparatively sluggish due to the gradient's exponential decline. At the same time, the complexity of the CNN network increases with more parameters, which results in longer training time for CNN models. Batch normalization and activation functions have significantly improved the reduction of vanishing gradients. However, optimizing CNN models with complex architecture remains challenging despite these advancements. These challenges motivate us to utilize ensemble frameworks for oral cancer detection.

Ensemble learning offers distinct advantages over single deep learning models by offering enhanced robustness, improved accuracy through model aggregation, diverse insights from multiple models, the ability to estimate uncertainty, effective management of data variability, and flexibility in combining models.

These advantages collectively motivate us to propose an ensemble framework for binary classification of histopathological oral specimen images.

B. CONTRIBUTION

The main contribution of this study is given below.

1) A novel framework is proposed for the classification of oral malignancy utilizing histopathological images. The proposed framework uses transfer learning and ensemble learning concepts.

2) The histopathological oral cancer image dataset is considered to evaluate the effectiveness of the proposed framework in the identification task of oral malignancy. Initially, these image undergoes preprocessing, where a Gaussian filter and Laplacian filter are applied to eliminate noise. Subsequently, color, contrast, and sharpness enhancements are performed.

3) As the used data set is imbalanced and to overcome the overfitting problem, image augmentation is applied, and an updated dataset is generated.

4) The proposed framework employs transfer learning with pre-trained CNN models in the initial phase for the classification of oral cancer dataset. To circumvent the constraint of any particular CNN model, six types of CNN are used with transfer learning in the experimental study, and the best two models, namely InceptionResNetV2 and Xception net, are considered as the base model for the suggested ensemble framework.

5) To prove the competence of the suggested work, its performance is evaluated and analyzed with other leading-edge models.

IV. MATERIALS AND METHODS

This study introduces a framework that integrates transfer learning and ensemble learning for classifying histopathological oral cell images into two categories. The proposed framework has utilized the concept of transfer learning and ensemble learning in two phases. In the first phase, transfer learning is applied to six types of CNN models selected for the classification task. Out of six models, the best two models are selected according to their superior performance. In the next phase, these two models are considered to form an ensemble model for the histopathological oral cell classification. The next two subsections discuss the methodology behind transfer learning and ensemble learning.

A. TRANSFER LEARNING

Medical imaging constitutes a robust resource within the data-driven healthcare landscape for diagnosing any type of disease, including cancer. Computer-aided diagnosis became popular with the advancement of ML and DL techniques. One of the significant hurdles in medical imaging is that it requires specialized equipment for capturing images and skilled physicians to label those images. As DL models require huge amounts of data, collecting sufficient training data became costly and strenuous. Therefore, the transfer learning concept can be applied to clinical image analysis. [36].

Transfer learning entails leveraging the intelligence acquired by a system from one problem to address a new and pertinent challenge. In this method, rather than initiating the training process from the beginning, the model's previously learned patterns are adapted to the fresh problem. Hence, the model's training time is decreased, and even with limited training data, the DL model can be trained. The pre-training of the DL models with publically available large datasets such as ImageNet and CIFAR is a standard research practice, and refining this model using examples of target domains with fewer parameters and smaller scales is a typical transfer learning technique [37], [38].

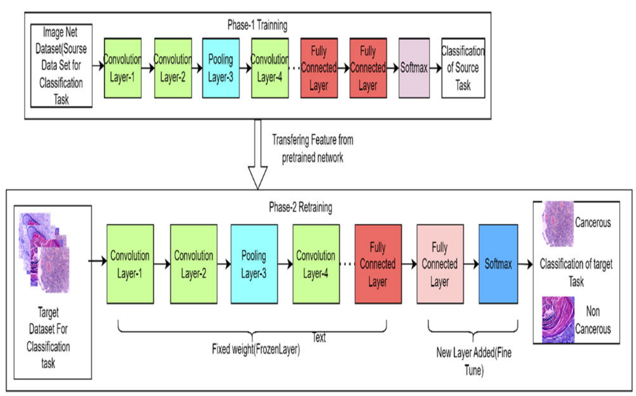

FIGURE 1. Concept of transfer learning.

If the source domain is given as, DSource = {Xsource, fsource(x)} where Xsource is the feature space, fsource(x) source predictive function, learning task is Tsource and the target domain is given as DTarget = {XTarget,, fTarget(x)}, where XTarget is the feature space of target domain, and fTarget(x) is the predictive function, TTarget is the learning task of the target domain, then transfer learning seeks to enhance the target predictive function fTarget(x) in DTarget using the knowledge of source domain DSource and source domain learning task Tsource, where DSource≠DTarge or Tsource≠TTarget. The general concept of transfer learning is illustrated in Fig. 1.

The workflow followed in the transfer learning process is given below

1) Use the layers of the pre-train model, which is previously trained in some other source domain.
2) Freeze those layers to prevent any data loss in the next training round.
3) Above the existing frozen layers, new trainable layers are appended.
4) On the target dataset, train the new layers of the pre-train model.
5) The model will discover how to predict using the previous feature.

Transfer learning is used in this research for subsequent purposes.

1) The data set available for the oral cancer histopathological image is relatively limited and is also expensive for medical experts to label the data. At the same time, the DL model needs a sufficient amount of annotated data. Thereby, transfer learning offers a solution to address the challenges of inadequate training data.
2) Most of the pre-process frameworks for the classification and identification of histopathological images are originated from ImageNet Large Scale Visual Recognition Challenge (ILSVRC). As their stability is already proven, these can be securely utilized for the oral cancer histopathological image classification task.
3) Transfer learning plays a significant role in enhancing accuracy and minimizing training duration.

B. ENSEMBLE LEARNING

Ensemble learning combines multiple models to address computational intelligence challenges by reducing variance and bias, thereby enhancing classification accuracy. A single model may not capture all dataset features adequately for accurate predictions; thus, merging different models retrieves additional features and improves prediction accuracy significantly. Ensemble models are typically created through methods like bagging, boosting, and stacking [39].

Bagging, or bootstrap aggregating, improves the stability and accuracy of machine learning algorithms in classification and regression tasks by training multiple base models on subsets of the original dataset and combining their predictions through model averaging [40].

In Boosting method an ensemble model is created by combining weak classifiers into a strong classifier. Initially, a model is trained on the entire dataset. Sequentially, additional models are built to correct errors made by previous models. This iterative process continues until the training dataset is accurately predicted or a maximum number of models is reached. Each model focuses on, correctly classifying misclassified data points by adjusting their weights iteratively, ensuring consistency across the dataset until the desired outcome is achieved [41].

Stacking is an ensemble method where models are combined based on their weights to create a new model. It operates in two layers: the first layer includes all models, while the second layer acts as the prediction layer. The dataset undergoes n-fold cross-validation, assigning each fold to a base model for predictions. This process repeats for all folds and models, generating level 0 predictions from the training data. These predictions are then used to train a level 1 model, whose output serves as features for the final model. The final model's performance is evaluated on the test dataset [42].

All the ways mentioned above of creating an ensemble model have some advantages and disadvantages. Bagging is mainly used to reduce the variance or decrease the overfitting issue, whereas boosting is used to reduce the bias or reduce the underfitting issue. Stacking is employed to enhance the classification or regression accuracy. If an ensemble model is created using bagging or boosting techniques, then the ensemble model's base models must be of a similar type. However, in stacking all the base models can be of heterogeneous type. In bagging, base models are executed parallelly, whereas in boosting base models are executed sequentially. In stacking, a meta-model is formed from the base model's features. The combination of base models is done in bagging using the max voting and averaging method. The aggregation of base models is done in boosting and stacking using the weighted average method [40], [41], [42]. All these differences are mentioned in Table 1.

TABLE 1. Difference between Bagging, Boosting, and Stacking Ensemble learning methods.

Ensemble Model	Bagging	Boosting	Stacking
Purpose of creation	Reduce Variance/Reduce Overfitting	Reduce Bias/Reduce Underfitting	Improve Accuracy
Base model Types	Homogeneous models	Homogeneous models	Heterogeneous models
Week or Strong Learner	Week Learner	Week Learner	Strong Learner
Base model execution method	Parallel execution	Sequential execution	Meta Model creation
Aggregation of base model method	Max Voting, Averaging	Weighted Averaging	Weighted Averaging

From the above discussed methods of ensemble model construction, in this current work, the stacking method is used to construct the suggested ensemble framework as this technique is a strong learner and accepts heterogeneous models as the base model.

C. PROPOSED ENSEMBLE CLASSIFIER FOR ORAL MALIGNANCY DETECTION(ECOMD)

In this current work, an ensemble classifier for oral malignancy detection (ECOMD) is proposed for the binary classification of oral histopathological images with the application of transfer learning and ensemble learning approaches. The algorithm for the suggested ECOMD is presented below.

Algorithm for ECOMD:
Input: Histopathological oral cell images
Output: Classified Histopathological oral cell images
Step 1: Dataset collection.
Dataset = CollectDataset()
Step 2: Image preprocessing.
PreprocessedImages = PreprocessImages(Dataset)
Step 3: Data Partition.
TrainningSet, ValidationSet, Testset =
PartitionData(PreprocessedImages)
Step 4: Image augmentation.
AugmentedTrainingSet = AugmentImages(TrainingSet)
Step 5: Transfer learning application to six selected CNN models.
TrainedModels =
ApplyTransferLearning(AugmentedTrainingSet)
Step 6: Base model selection for ensemble model out of six CNN models used in the previous step.
BaseModel1, BaseModel2 =
SelectBaseModels(TrainedModels)
Step 7: Construction of Ensemble model using two selected base models.
EnsembleModel =
ConstructEnsembleModel(BaseModel1, BaseModel2)
Step 8: Classification of oral images using the constructed ensemble model.
ClassifiedImages =
ClassifyImages(EnsembleModel, TestSet)
Step 9: Performance evaluation of ensemble model.
PerformanceMetrics =
EvaluatePerformance(ClassifiedImages)

The proposed ECOMD consists of nine steps, as given in the above algorithm. In step 1, data collection is done; in step 2, image preprocessing is carried out; in step 3, data partition takes place. As the number of images is fewer, in step 4, new images are generated using different image augmentation techniques. In step 5, CNN models are utilized to classify the dataset using a transfer learning application. Step 6 selects two base models from the previous step based on their performance. Construction of an ensemble model using the two selected base models is done in step 7. This ensemble model is then tested for the oral cancer classification task in step 8. Finally, in step 9, the effectiveness assessment or performance evaluation of the ensemble model is carried out utilizing different performance indicators. Fig. 2 illustrates the flow of 9 steps of the proposed ECOMD.

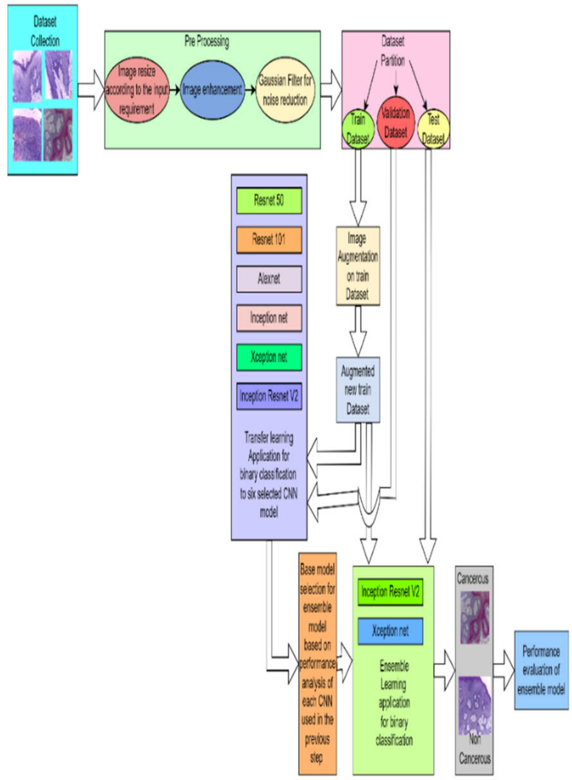

FIGURE 2. Proposed model for binary classification of oral cancer image using ensemble learning.

V. METHODOLOGIES

In this section, the methodologies used for the proposed ECOMD are discussed. As the considered framework consists of 9 steps, the concept behind each step is discussed in the following 9 subsections.

A. DATASET USED

The used data set is published by the author Rahman et al. [43] and can be accessed openly. The images of two different magnifications, of histopathological oral cells are obtainable in the dataset. Specifically, 439 cancerous and 89 normal images are of 100 $\times$ magnification. Similarly, 495 cancerous and 201 noncancerous images are of 400 $\times$ magnification. A total of 1224 histopathological oral cell images are available in this image dataset. As this is an image dataset, before using this dataset in the proposed model, image preprocessing is applied to decrease the noise in the images. The current dataset suffers from imbalance and lacks sufficient images, both of which are crucial for enhancing the performance of deep learning models. Research shows that balanced datasets with a larger number of samples tend to yield better results in deep learning applications. Table 2 represents the information regarding the images of the dataset. Table 2 shows that out of the 1224 images available, 934 images are of cancerous cells, and the remaining 290 are noncancerous images.

TABLE 2. Details of images available in the used dataset.

Type of image in the used Data set	Count of Cancerous images	Count of Noncancerous images	Total
100 × magnification	439	89	528
400 × magnification	495	201	696
	Total= 934	Total=290	1224

It can be noticed that the total count of images in the dataset is less and unbalanced, so image augmentation is applied after the preprocessing step to make the dataset balance and generate a larger dataset out of the original dataset. Fig. 3 shows some samples of images used.

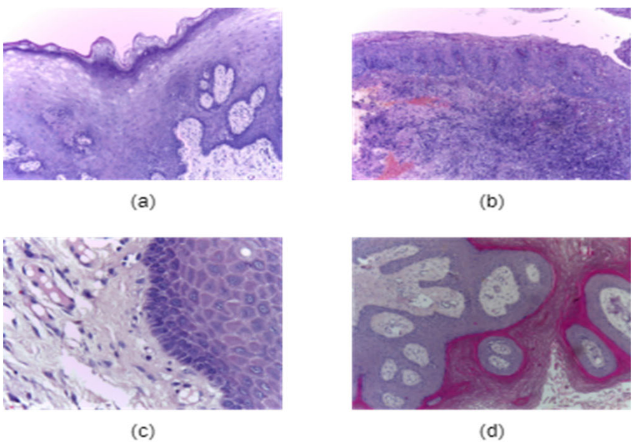

FIGURE 3. Sample images of the dataset. (a) Normal image of 100 × magnification. (b) Cancerous image of 100 × magnification. (c) Normal image of 400 × magnification. (d) Cancerous image of 400 × magnification.

B. IMAGE PREPROCESSING

The images in the utilized dataset come in various sizes, so in the preprocessing stage, all the images are resized to a particular size according to the input requirement of the model. In the preprocessing stage, image enhancement is carried out using colour, contrast, and sharpness. After image enhancement, the Gaussian filter is applied to reduce noise. This lowpass filter is used to smoothen the images used in this work. Fig. 4 shows an example of the image before and after the preprocessing stage.

C. DATASET PARTITION

After the image preprocessing step, the dataset is partitioned with a distribution of 80% for training, 10% for validation, and 10% for testing. As the total available images is 1224,

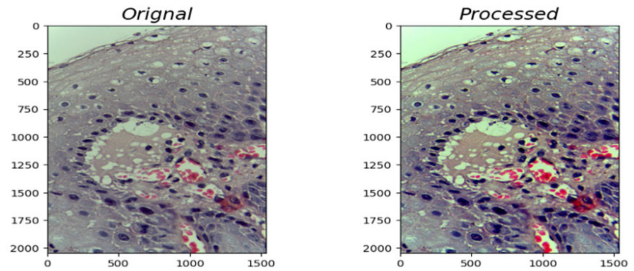

FIGURE 4. Example of original and processed images.

the training set holds 980, and the validation and testing set individually contain 122 and 122 images, respectively. The specific dataset partition information is given in Table 3. As the total count of images available in the training set is smaller and imbalanced, and training DL models demand a significant volume of data, before using this dataset, a new augmented training dataset is generated from the existing dataset.

TABLE 3. Dataset partition information.

	Total count of training set images = 980		Total count of validation set images = 122		Total count of test set images = 122	
Type of image in the Dataset	Total count of Cancerous images in training set	Total count of Non cancerous images in training set	Total count of Cancerous images in validation set	Total count of Non cancerous images in validation set	Total count of Cancerous images in testing set	Total count of Non Cancerous images in testing set
100 × magnification	352	72	44	9	43	8
400 × magnifi	396	160	49	20	50	21
	Total= 748	Total = 232	Total= 93	Total = 29	Total= 93	Total= 29

D. IMAGE AUGMENTATION

DL models necessitate an extensive dataset for training, enhancing their performance with larger, well-balanced training sets [44]. In this work, the dataset consists of only 1224 images of imbalanced cancerous and non-cancerous oral histopathological images. Image augmentation is applied to the existing training dataset to make the training dataset balanced and create a big training dataset. As discussed in the above data partition section, out of 1224 images, 980 images are kept in the training set. Different type of image augmentation, such as rotation, flipping, tilting, random distortion, zooming, shirring, and cropping in the center of the image, is applied to these 980 images, and another 5170 new augmented images are generated. Together, in the new training set, 6150 images are kept. Fig. 5 represents one of the original OSCC image with its corresponding augmented image generated through zooming. Similarly, Fig. 6, Fig. 7 and Fig. 8 show some examples of augmented images generated through elastic distortion, center cropped and affine transformation respectively. The advantage of image augmentation is that it creates new images from the existing ones without losing any features from the existing images. The new training set contains balanced cancerous and non-cancerous images. Table 4 shows the details of augmented image-generated information.

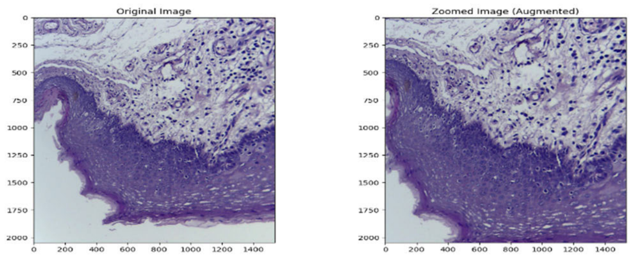

FIGURE 5. Example of original and augmented zoomed image.

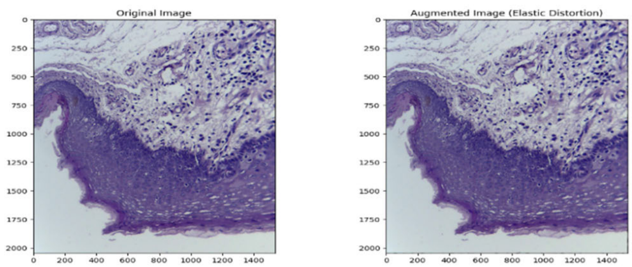

FIGURE 6. Example of original and augmented elastic distortion image.

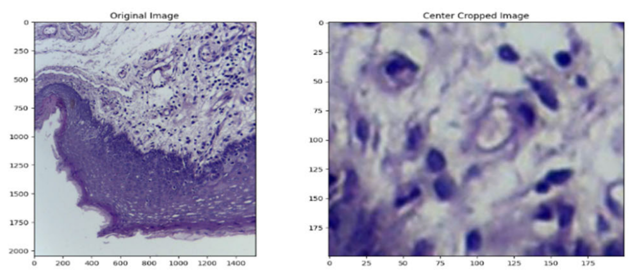

FIGURE 7. Example of original and augmented center cropped image.

E. TRANSFER LEARNING APPLICATION TO SIX CNN MODEL

In the next step after the image augmentation step, transfer learning is applied to six selected CNN models. In this work, Inception, Alexnet and ResNet series are carefully analyzed, and six CNNs, namely Alexnet, Inception net, ResNet50, ResNet101, InceptionresnetV2 and Xception net, are selected

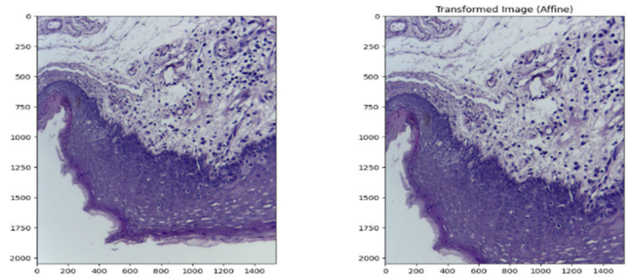

FIGURE 8. Example of original and affine transformed image.

TABLE 4. Augmented image generation information.

Type of image in training set	Count of cancerous images prior to augmentation in the training set	Count of Non cancerous images prior to augmentation in the training set	Count of Cancerous images post augmentation in the training set	Count of Noncancerous images post augmentation in the training set
100 × magnific	352	72	1400	1000
400 × magnific	396	160	1675	2075
	Total=748	Total= 232	Total= 3075	Total= 3075
	Total=748+232=980		Total= 3075+3075=6150	

for binary classification of oral malignancy using histopathological images. Transfer learning is applied to these selected CNNs. Firstly, labelled images from the ImageNet data set are applied to each individually selected model, then the front layers of the selected CNNs are frozen, and new and fresh trainable layers are added. The new layers are trained using the augmented histopathological oral images. Ultimately, the chosen CNN's effectiveness was assessed using the validation dataset.

F. BASE MODEL SELECTION FOR ENSEMBLE MODEL

In this step, based on the performance, out of six CNN models used in the previous step of the transfer learning application, only two CNNs, namely Xception net and Inceptionresnet V2, are considered for the comprehensive ensemble model.

G. CONSTRUCTION OF ENSEMBLE MODEL

In this work, the stacking method discussed in section IV (subsection B) is used to construct an ensemble model as the purpose is to increase the classification accuracy. For the comprehensive ensemble model, two CNN models, namely InceptionResnetV2 and Xception net, are used based on better performance out of the six CNN models used in the experiment.

H. CLASSIFICATION OF ORAL IMAGES USING ENSEMBLE MODEL

The ensemble model constructed in the previous step is applied for the classification of the histopathological oral images. The ensemble model is trained using the augmented train dataset.

I. PERFORMANCE EVALUATION

In the final step, effectiveness assessment or performance evaluation of the suggested ensemble approach is done by applying the test dataset. Four different statistical measures are used for the performance evaluation: precision, recall or sensitivity, f1 score, and accuracy. In this study, we use precision to determine how accurately a sample is identified as cancerous, is truly cancerous. Recall or sensitivity is used to measure, out of all the cancerous samples, how many samples are properly identified as cancerous. For enhancing recall and precision, the F1 score is used; it is the harmonic mean of recall and precision. Accuracy is used to measure how often the model's prediction is correct. Accuracy is the ratio of the sum of correct predictions to the whole number of predictions [45]. Equations 1 to 4 give the mathematical notation of each statistical measure used. In the equations, TP indicates True positive, FP is false positive, FN is false negative, and TN is True negative.

$$\text{Precision} = \frac{\text{TP Ratio}}{\text{TP Ratio} + \text{FP Ratio}} \tag{1}$$

$$\text{Recall or Sensitivity} = \frac{\text{TP Ratio}}{\text{TP Ratio} + \text{FN Ratio}} \tag{2}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

$$\text{Accuracy} = \frac{\text{TP Ratio} + \text{TN Ratio}}{\text{TP Ratio} + \text{FN Ratio} + \text{FP Ratio} + \text{TN Ratio}} \tag{4}$$

VI. EXPERIMENTAL ENVIRONMENT

The proposed framework, discussed in the above nine steps, is executed using Python in the Google Colab platform. All the used six CNN models, namely Inception net, Alexnet, ResNet50, ResNet101, InceptionresnetV2 and Xception, are implemented with a transfer learning application on the graphics processing unit(GPU), available in the Google Colab platform. Along with that, the proposed ensemble model has also been implemented on the Google Colab platform. All the DL models used in this research were implemented using the Keras framework, and the TensorFlow framework is used in the backend of Keras.

A. HYPERPARAMETER TUNING

Within this segment, the best hyperparameter is set for the suggested framework. Firstly, the hyperparameters are set for the six CNN models used in this study. After various trials with different combinations of hyperparameters, the six CNNs are trained over 100 epochs, and the idle parameter is set. Particularly, the Adam optimizer is used during the model training procedure. Three different learning rates (0.01.0.001,0.0001) are explored, and three different batch sizes (16,32,64) are also explored. Various dropouts (0.5,0.4,0.3,0.2,0.1) are also explored to overcome the overfitted problem during training. The optimal configuration, noted to reduce the disparity between training loss and validation loss, includes a learning rate of 0.001, a batch size of 32, and a dropout rate of 0.2. Finally, two models, namely Xception Net and InceptionResNet V2, have been chosen based on their performance metrics, exhibiting lower validation loss and higher validation accuracy. The weights of those two models are saved and used in the ensemble framework for the histopathological image classification in the test dataset. The optimum hyperparameter set for the Xception net and Inceptionresnet V2 is given in Table 5.

TABLE 5. Optimum hyperparameter set for Xception net and Inceptionnet V2.

Hyperparameter	Xception net	Inceptionresnet V2
Kernel size	5×5	5×5
Dropout Rate	0.05	0.05
Learning Rate	0.001	0.001
Batch Size	32	32
Input image size	299×299	299×299
Optimizer	Adam	Adam

VII. RESULT ANALYSIS

The result analysis of the suggested model is discussed and analyzed in three different ways, (i) assessing performance through statistical metrics, (ii)assessing performance using learning curves(iii) performance evaluation using comparative study with other models from literature.

A. ASSESSING PERFORMANCE THROUGH STATISTICAL METRICS

The effectiveness or performance of all the six CNN models used, and the suggested ensemble model is discussed in this section. The statistical measures are calculated for all the considered six CNNs and for the suggested ensemble framework. The statistical measures are calculated using equations 1 to 4 given in section V(subsection I. Performance Evalution) and recorded in Table 6. It can be noticed that, out of six CNN models, Xception net and Inceptionresnet V2 give the best results with 95% and 93% accuracy, respectively. Based on the performance these two models are set for the next level ensemble model. It can also be apparent that the suggested ensemble model provides comparatively the best outcome with 99% precision, 93% recall, 98% f1 score, and 98% accuracy.

These metrics indicate a high-performing model overall, with very high precision and accuracy. Here's how we can interpret the failure analysis from these numbers:

Precision (99%): This suggests that when the model predicts a positive class, it is correct 99% of the time. In other words, it makes very few false positive errors.

Recall (93%): The recall indicates that the model correctly identifies 93% of all actual positive instances. While this is quite high, it means that the model misses about 7% of positive cases, suggesting there are some false negatives.

F1 Score (98%): The F1 score, which is the harmonic mean of precision and recall, also reflects the high performance of the model in terms of balancing precision and recall. A score of 98% indicates strong overall performance in classification tasks.

While the metrics suggest the model performs well overall, a comprehensive failure analysis entails investigating potential weaknesses such as overlooking positive instances (false negatives), its ability to generalize to new data, issues related to data quality, optimization of classification thresholds, and understanding the rationale behind the model's decisions.

TABLE 6. Performance analysis using statistical measures.

Model Name	Precision	Recall	F1-Score	Accuracy
Resnet 50	0.91	0.92	0.91	0.91
Resnet101	0.89	0.90	0.90	0.89
Alexnet	0.88	0.89	0.89	0.88
Inception Net	0.92	0.92	0.92	0.92
Xception Net (Selected Base model 1)	0.93	0.96	0.94	**0.95**
Inceptionresnet V2 (Selected Base model 2)	0.93	0.93	0.93	**0.93**
Proposed Ensemble model	**0.99**	**0.93**	**0.98**	**0.98**

B. PERFORMANCE EVALUATION USING LEARNING CURVES

This section discusses the learning curves, particularly accuracy and loss curves, for performance analysis purposes. We consider the loss and accuracy curves for both training and validation sets. The models are trained with 100 epochs, and the curves are recorded for analysis purposes.

Fig. 9 shows the learning curves of the Resnet 50 network. From the figure, it is apparent that the intermediary distance

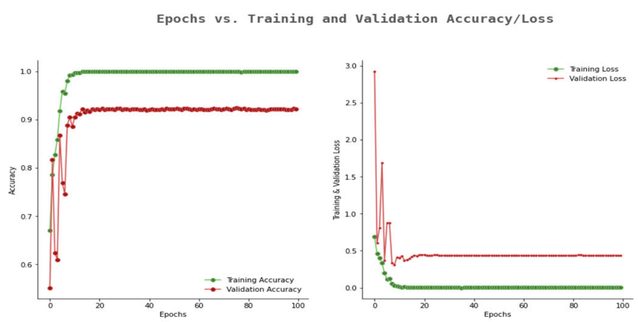

FIGURE 9. Accuracy and loss curves of Resnet 50.

of the validation and training accuracy curve is coming to a constant after the 20th epoch, and similarly, the intermediary distance of the training and validation loss curve also comes to a constant after the 20th epoch, which shows the Resnet50 model does not get the problem of overfitting.

The accuracy and loss curve of Resnet 101 is presented in Fig. 10. It is apparent from the figure the training and validation accuracy graph is coming to a constant after the 20th epoch. The maximum validation accuracy attended by this model is 0.89, whereas the maximum training accuracy is 0.99. It can also be observed that the distance between training and validation accuracy is also greater as compared to other models considered.

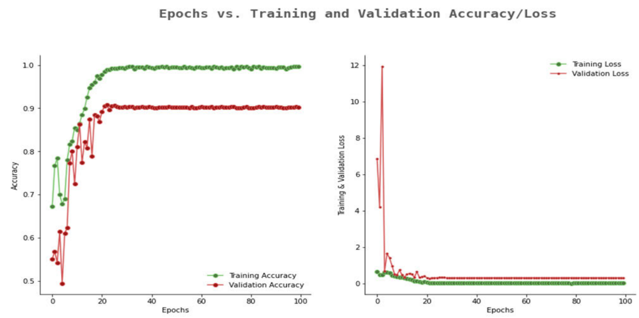

FIGURE 10. Accuracy and loss curves of Resnet 101.

Fig. 11 shows the curves of the Alexnet model, and it is apparent that the intermediary distance of training and validation accuracy, as well as the discrepancy in training and validation loss, is smaller than that of the Resnet50 and Resnet101 models. Though the model minimizes the generalization disparity of training and validation accuracy, the model reached a maximum accuracy of 0.88, which is comparatively lower than the previous two models.

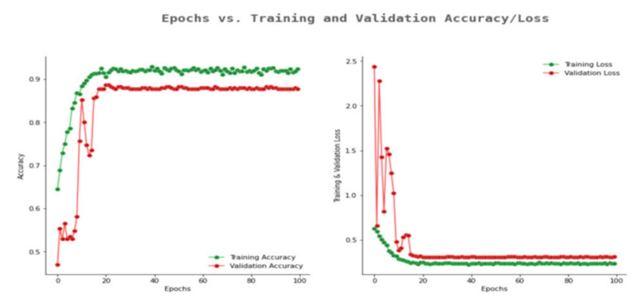

FIGURE 11. Accuracy and loss curves of Alexnet.

Fig. 12 shows the curves of the Inception net; it achieved a maximum testing accuracy of 0.92 and training accuracy of 0.99, whereas the training and validation loss achieved a constant value of 0.01 after 3rd epoch.

Fig. 13, provided curves of the Xception network. Analysis of the curve reveals that the Xception model attained a peak testing accuracy 0.95 and a training accuracy 0.99. The validation loss curve also comes to a constant value of 0.2 after the 10th epoch.

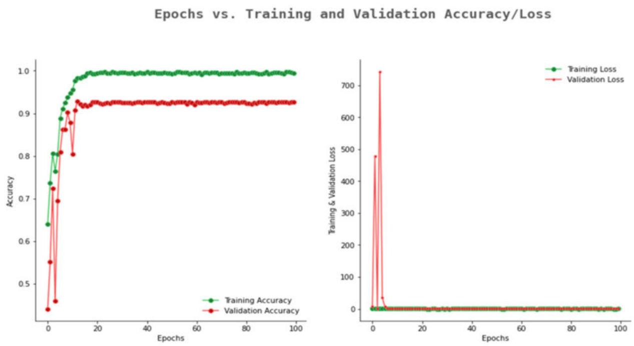

FIGURE 12. Accuracy and loss curves of inception net.

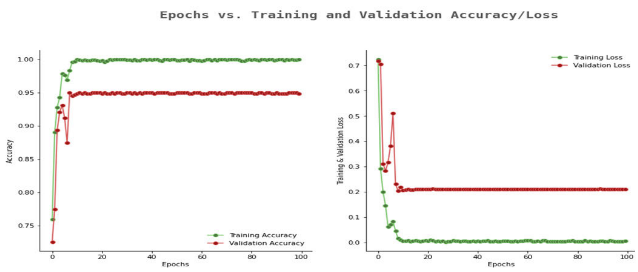

FIGURE 13. Accuracy and loss curves of Xception net.

In Fig. 14, curves of Inceptionresnet V2 are shown. From the curves, it can be observed that this model minimizes the generalization gap of training and validation, both in accuracy and loss curve recorded within 100 epochs. This model reached an accuracy of 0.93.

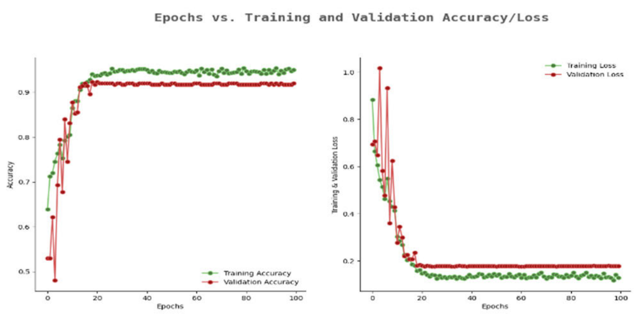

FIGURE 14. Accuracy and loss curves of Inceptionresnet V2.

From all the above-discussed learning curves, the best-generalized curve generated are for the model Xception net and Inceptionresnet V2 with an accuracy of 0.95 and 0.93 respectively. These two networks are used as the base models for the ensemble model.

Fig. 15 represents the curves of the suggested ensemble model. It is noticeable from the figure that the ensemble model reaches a constant accuracy of 0.98 after the 17th epoch and the disparity of the training and validation accuracy is also minimized.

At the same time, the generalization gap between the training and validation loss curve is also minimized with a constant validation loss after the 17th epoch.

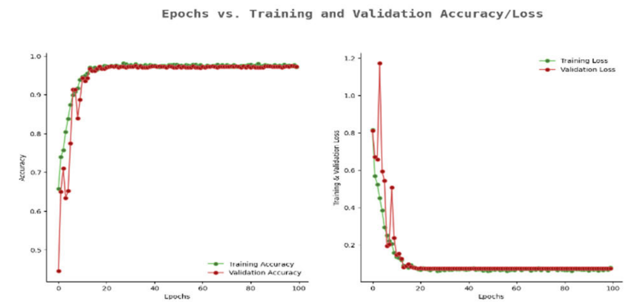

FIGURE 15. Accuracy and loss curves of the proposed ensemble model.

From the above observation, it can be concluded that, out of six CNN models used in the experiment, the Xception net and Inceptionresnet V2 models are performing better results with 0.95 and 0.93 validation accuracy, respectively, whereas the ensemble model with the Xception net and Inceptionresnet V2 as the base model still able to increase the effectiveness or performance in terms of validation accuracy. The proposed ensemble model got the highest accuracy recorded at 0.98 compared to the other six CNN models, which are implemented with transfer learning applications.

C. COMPARATIVE ANALYSIS WITH THE EXISTING LITERATURE

Here, the effectiveness in terms of accuracy achieved by the suggested ensemble approach is measured against the

TABLE 7. Comparative analysis with available literature.

Model from literature	Method used	Type of images used	Accuracy in %
Dev kumar Das et al. [29]	CNN model	Histopathological oral image	96.88
Ibrar Amin et al. [30]	Concatenated DL model	Histopathological oral image	96.66
Navarun et al. [31]	CNN model	Histopathological oral image	97.5
Santisudha et al. [32]	Modified Resnet model	Histopathological oral image	97.59
Santisudha et al. [33]	Modified capsule network	Histopathological oral image	97.35
Madhusmita Das et al. [34]	CNN model	Histopathological oral image	97.82
Sukegawa et al.[35]	CNN model	Histopathological oral image	86.22
Proposed Ensemble model	Ensemble model	Histopathological oral image	97.88

accuracy of the other model available in the literature. Here, we considered only those models that have used histopathological oral images for the classification task of oral malignancy, employing DL methods. Table 7 indicates some of the models from the literature for the comparative analysis.

It is evident from the preceding Table 7 that the suggested ensemble framework's performance in terms of accuracy is comparable with the other models from the literature. The suggested model got an improved accuracy of 97.88% as compared to other models for the oral cancer classification task.

VIII. CONCLUSION & FUTURE SCOPE

Oral cancer classification using histopathological images is a challenging and rigorous task. In this work, we compare and analyze various CNN models with transfer learning applications and finalize the best-performed CNN models (Xception net and Inceptionresnet V2) as the base models for the construction of the suggested ensemble model. As the CNN models used for the oral cancer classification task do not undergo the problem of overfitting or underfitting, the main aim of the suggested ensemble model is to try to enhance the performance in terms of accuracy. In this work, the stacking method is applied for the construction of an ensemble model, and the model achieves a better accuracy of 97.88% as compared to base models. The performance or effectiveness of the model in terms of accuracy is also compared and analyzed with other models available in the literature, and it was found that the results are comparable with other models. In further work, the suggested model can be tested to identify different phases of oral cancer using histopathological oral images.

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